Tafasitamab, lenalidomide, and rituximab in relapsed or refractory follicular lymphoma (inMIND): a global, phase 3, randomised controlled trial

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Summary

Background Follicular lymphoma is characterised by episodes of remission and relapse, with patients requiring multiple lines of therapy. Lenalidomide plus rituximab is a commonly used immunotherapy combination in patients with relapsed or refractory follicular lymphoma. We aimed to assess the efficacy and safety of adding tafasitamab, a CD19-targeted Fc-enhanced monoclonal antibody, to lenalidomide and rituximab in this setting.

Methods This phase 3, double-blind, randomised, placebo-controlled trial (inMIND) was done in 210 centres (including community-based haematology clinics, major hospitals, and academic institutions) in North America, Europe, and the Asia–Pacific region. Adults with relapsed or refractory follicular lymphoma, who had received at least one previous line of systemic therapy, were eligible for enrolment and randomly assigned (1:1) to receive treatment with up to 12 cycles (28-day cycle length) of tafasitamab (12 mg/kg by intravenous infusion on days 1, 8, 15, and 22 of cycles 1–3 and days 1 and 15 of cycles 4–12) or placebo, both with lenalidomide (20 mg/day orally on days 1–21 of cycles 1–12) and rituximab (375 mg/m² by intravenous infusion on days 1, 8, 15, and 22 of cycle 1 and day 1 of cycles 2–5). Treatment assignment was achieved via an interactive voice or web response system; patients, investigators, and the funder were masked until the primary analysis. Study endpoints were investigator assessed unless otherwise specified. The primary endpoint was progression-free survival in the intention-to-treat population of all randomised patients; safety was assessed in all randomised patients who received at least one dose of study drug. The trial is registered with ClinicalTrials.gov (NCT04680052) and EUDRA-CT (2020–004407–13) and is active but no longer enrolling.

Findings Between April 16, 2021, and Aug 10, 2023, a total of 817 patients were assessed for eligibility; 548 patients with relapsed or refractory follicular lymphoma were enrolled and randomly assigned to treatment with either tafasitamab (n=273) or placebo (n=275). 299 (55%) of all randomised patients were male and 249 (45%) were female. The addition of tafasitamab to lenalidomide and rituximab resulted in significantly lower risk of progression, relapse, or death versus placebo (median progression-free survival by investigator 22·4 months [95% CI 19·2 to not evaluable] in the tafasitamab group *vs* 13·9 months [11·5–16·4] in the placebo group; hazard ratio 0·43 [95% CI 0·32–0·58]; p<0·0001) in the planned primary analysis. Improvement in progression-free survival was confirmed by independent review committee. Adverse events were reported in 272 (99%) of 274 patients in the tafasitamab group and 270 (99%) of 272 patients in the placebo group. Most common adverse events occurring in either the tafasitamab group or placebo group were neutropenia (133 [49%] *vs* 123 [45%]) and diarrhoea (103 [38%] *vs* 77 [28%]). There were no deaths due to treatment-related adverse events in the tafasitamab group; two (1%) patients had fatal adverse events related to treatment in the placebo group.

Interpretation The addition of tafasitamab to lenalidomide and rituximab resulted in a statistically significant and clinically meaningful improvement in progression-free survival, with an acceptable safety profile in patients with relapsed or refractory follicular lymphoma. This combination represents a potential new standard-of-care treatment.

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Introduction

Follicular lymphoma is the most common indolent non-Hodgkin lymphoma, comprising 20–25% of all non-Hodgkin lymphoma cases in high-income countries.[1,2] Although follicular lymphoma is associated with high rates of long-term survival (10-year overall survival 77–85%), it is generally incurable and characterised by remissions and relapses.[3,4] In the majority of cases, first-line therapy for patients with follicular lymphoma relies on chemotherapy in

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See Online for appendix

Research in context

Evidence before this study

To more fully understand the current treatment landscape and associated challenges in indolent non-Hodgkin lymphoma, we searched the PubMed database for articles published in English between Jan 1, 2020, and March 10, 2025, on the role of the tumour microenvironment as well as the impact of differences in CD20 expression in tumours on disease progression and development of treatment resistance. We used the search terms "((“non-Hodgkin lymphoma” OR “follicular lymphoma” OR “marginal zone lymphoma”) AND CD20) AND English [Language] AND (“2020/01/01”[Date - Publication]: “2025/03/10”[Date - Publication])”. Based on published reports from the search, the tumour microenvironment restricts host immune responses and treatment with anti-CD20 therapy (eg, rituximab) might reduce CD20 expression on tumour cells, contributing to resistance or relapse after monoclonal antibody therapy. Further, prognosis generally worsens with each subsequent therapy in patients with relapsed or refractory disease, highlighting the need for the development of additional treatment options that could further improve survival outcomes. Tafasitamab is an Fc-enhanced humanised monoclonal antibody that was designed to target the broadly expressed B-cell antigen CD19. In preclinical studies in models of diffuse large B-cell lymphoma and Burkitt lymphoma, antitumour activity was increased compared with respective monotherapy treatments when tafasitamab was combined with rituximab and different chemotherapeutic and non-chemotherapeutic drugs, including lenalidomide. Tafasitamab is currently approved in the USA, the EU, and other countries in combination with lenalidomide for the treatment of adult patients with relapsed or refractory, diffuse, large B-cell lymphoma who are not eligible for autologous stem cell transplantation.

Added value of this study

In a follicular lymphoma patient population with high burden of disease, including many who were refractory to anti-CD20 monoclonal antibody therapy, tafasitamab in combination with lenalidomide and rituximab was associated with significantly longer progression-free survival compared with placebo added to lenalidomide and rituximab. This improvement was seen in all subgroups analysed (including in patients with disease progression or relapse within 24 months of initial diagnosis status, anti-CD20-refractory status, and by number of previous lines of therapy). Additionally, tafasitamab was associated with higher rates of overall response and metabolic complete response versus placebo. The safety profile of this combination was consistent with expectations and manageable for this population, and, importantly, the addition of tafasitamab did not compromise lenalidomide or rituximab delivery. To our knowledge, our study is the first to validate the approach of combining two unconjugated monoclonal antibodies (anti-CD19 with anti-CD20) in treatment of lymphoma in a large, phase 3, randomised clinical trial.

Implications of all the available evidence

CD19 represents a potential treatment target for patients with relapsed or refractory follicular lymphoma following anti-CD20 therapy. Data from our study demonstrate promising results of the addition of tafasitamab to lenalidomide and rituximab. Because this combination can be administered in the community as well as tertiary centres, these encouraging results indicate that it might become a new option for standard-of-care treatment for patients with relapsed or refractory follicular lymphoma, particularly as a second-line treatment for which few options are currently available. Future analyses will yield robust data on overall survival, as well as a more in-depth examination of quality of life in patients treated with tafasitamab. Additionally, analyses of CD19 expression in patients following treatment with tafasitamab will provide information to help guide subsequent treatment decisions.

combination with an anti-CD20 monoclonal antibody, which results in prolonged progression-free survival.[5–7] However, chemo-immunotherapy can be associated with substantial toxicity and development of disease resistance, yielding shorter durations of response when administered as subsequent lines of therapy.[8–10] As such, immunotherapy has become the preferred approach for patients with relapsed or refractory disease.[11,12] Based on the AUGMENT randomised study,[13] the immunomodulatory drug lenalidomide combined with rituximab is approved for the treatment of patients with follicular lymphoma after at least one previous line of therapy, and is commonly used.[14]

CD19 is a key mediator of B-cell proliferation and signalling that is broadly expressed during B-cell development and in most B-cell malignancies, including follicular lymphoma.[15,16] Tafasitamab, a CD19-targeted, Fc-enhanced, humanised monoclonal antibody, induces direct cytotoxicity and enhances natural killer cell and macrophage immune-mediated mechanisms.[17–20] Preclinical evidence suggests the potential to enhance outcomes in combination with the anti-CD20 monoclonal antibody rituximab[20] in patients with lymphoma. Tafasitamab in combination with lenalidomide is approved for the treatment of adult patients with relapsed or refractory, diffuse, large B-cell lymphoma who are ineligible for autologous stem-cell transplantation, based on results from the L-MIND study.[21–24]

inMIND was designed to evaluate the efficacy and safety of tafasitamab in combination with lenalidomide and rituximab versus placebo plus lenalidomide and rituximab in patients with relapsed or refractory follicular lymphoma or marginal zone lymphoma.

Methods

Study design

inMIND was an international, phase 3, multicentre, double-blind, randomised, placebo-controlled study. It was conducted at 210 centres (including community-based haematology clinics, major hospitals, and academic institutions) in North America, Europe, and the Asia–Pacific region (see appendix pp 2–5 for full list of study investigators and sites). Study sites were selected with personnel experienced in managing and treating the study patient population, and in conducting clinical trials. The trial protocol and statistical analysis plan (appendix p 15) was approved by local institutional review boards or ethics committees; the protocol and ethics approval for BC Cancer (LHS) was provided by the University of British Columbia BC Cancer Research Ethics Board (REB number: H21–00563), and for the Charles University (MT) by the European Medicines Agency European Clinical trial Register (EU CT number: 2023–504684–16–00). The study was conducted in accordance with the Declaration of Helsinki, Good Clinical Practices, and applicable local laws and specific regulations. The trial is registered with ClinicalTrials.gov (NCT04680052) and EUDRA-CT (2020–004407–13) and is active but no longer enrolling. An independent data monitoring committee monitored safety throughout the study. Various stakeholders, including healthcare professionals and regulatory agencies, provided input on the study protocol.

Patients

Eligible patients were aged at least 18 years, with histologically confirmed CD19$^+$ and CD20$^+$ follicular lymphoma (grade 1/2 or 3A) and documented relapsed or refractory disease after at least one previous line of systemic therapy, including an anti-CD20 monoclonal antibody (at least four doses), and an Eastern Cooperative Oncology Group performance status of less than or equal to 2. Patients also had to have measurable disease identified radiologically. Patients had to be in need of treatment for relapsed or refractory disease; Groupe d'Etude des Lymphomes Folliculaires (GELF) criteria were provided in the protocol as guidance, but ultimately the decision regarding the need for treatment was based on the discretion of investigators. In the case of disease progression within 24 months of initial diagnosis (POD24), new biopsies were required to exclude transformation to aggressive lymphoma. Patients were excluded if they had biopsy-proven or clinical evidence of transformed lymphoma by investigator assessment. A full list of eligibility criteria is provided in the protocol. Sex (male or female), race, and ethnicity data were self-reported by patients enrolled in the study. All patients provided written informed consent.

Randomisation and masking

Patients were randomly assigned (1:1) to receive tafasitamab or placebo plus lenalidomide and rituximab, and were stratified by POD24 status, refractoriness to previous anti-CD20 monoclonal antibody therapy, and number of previous lines of therapy (1 *vs* ≥2). Patients were centrally assigned to treatment groups using an interactive voice or web response system to conceal allocation. Patients, investigators, and the funder remained masked to each patient's treatment assignment until the primary analysis for the study. Tafasitamab and placebo infusions were prepared in saline bags; the infusion bag and lines were covered to block the solution colour to ensure treatment was not revealed during infusion and the entire duration of the study.

Procedures

Full details of the schedule of assessments and visits can be found in the protocol (appendix p 15). In brief, patients received up to 12 cycles (28-day cycle length) of tafasitamab

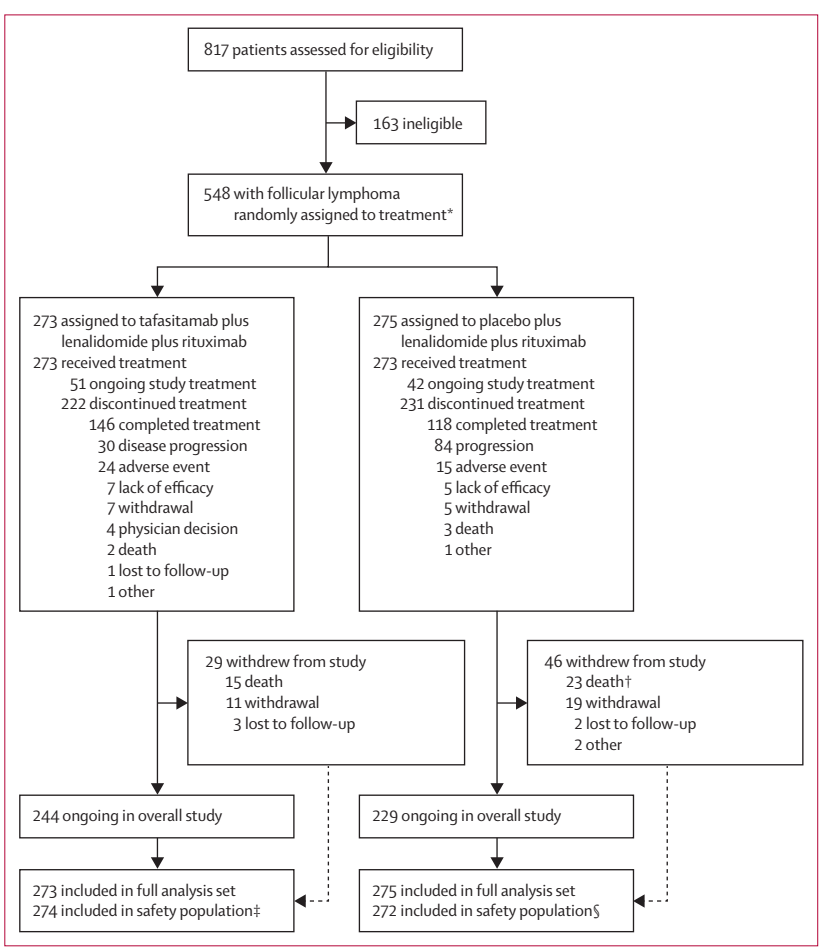

Figure 1: **Patient disposition**

Data are presented for the intention-to-treat population (full analysis set) and safety populations as of the data cutoff date of Feb 23, 2024. *106 patients had marginal zone lymphoma (not described in this report). †Death for one patient was reported but not recorded in the end-of-study form. ‡One patient randomly assigned to the placebo plus lenalidomide and rituximab group is included in the tafasitamab plus lenalidomide and rituximab safety population because the patient erroneously received tafasitamab. §Three patients randomly assigned to the placebo plus lenalidomide and rituximab group are not included in the safety population because they erroneously received tafasitamab (n=1) or did not receive any study treatment due to confirmation of rituximab hypersensitivity (n=1), and withdrawal by patient (n=1).

(12 mg/kg by intravenous infusion on days 1, 8, 15, and 22 of cycles 1–3 and days 1 and 15 of cycles 4–12) or placebo, combined with lenalidomide (20 mg/day orally on days 1–21 of cycles 1–12) and rituximab (375 mg/m² by intravenous infusion on days 1, 8, 15, and 22 of cycle 1 and day 1 of cycles 2–5); infusions were administered by qualified site personnel. CT scans were done at baseline and then for response assessment and surveillance every 3 months in year 1, then every 4 months in years 2 and 3, and every 6 months in years 4–6. PET–CT scans were done at baseline and repeated in the fluorodeoxyglucose (FDG)-avid population to confirm radiological complete response within 28 days, or at completion of therapy if radiological complete response was not confirmed previously. Any protocol deviation was categorised as important or not important. This was assessed by the funder using protocol-specific adjudication criteria to assess the effect of the deviation. All reported protocol deviations were deemed not significant as they did not affect assessment of outcomes and the overall findings from the study. Trial site investigators were responsible for data collection.

Adverse events were coded using Medical Dictionary for Regulatory Activities version 26.0 and severity of events was assessed using National Cancer Institute Common Terminology Criteria for Adverse Events version 5.0. Adverse events were reported by qualified site personnel and assessed for relationship to study treatment at each study visit and on an ongoing basis from initial treatment to the end of the safety follow-up period (90 days after last study treatment).

	Tafasitamab plus lenalidomide plus rituximab (n=273)	Placebo plus lenalidomide plus rituximab (n=275)	Total (N=548)
Age, years	64·0 (36–88)	64·0 (31–85)	64·0 (31–88)
≥65 years	136 (50%)	136 (50%)	272 (50%)
≥75 years	54 (20%)	54 (20%)	108 (20%)
Sex			
Male	150 (55%)	149 (54%)	299 (55%)
Female	123 (45%)	126 (46%)	249 (45%)
Ethnicity			
Hispanic or Latino	31 (11%)	24 (9%)	55 (10%)
Not Hispanic or Latino	228 (84%)	226 (82%)	454 (83%)
Not reported	11 (4%)	23 (8%)	34 (6%)
Unknown	3 (1%)	2 (1%)	5 (1%)
Race			
White	219 (80%)	219 (80%)	438 (80%)
Black or African American	1 (<1%)	0	1 (<1%)
Asian	40 (15%)	42 (15%)	82 (15%)
American Indian or Alaska Native	0	0	0
Native Hawaiian or Pacific Islander	0	0	0
Not reported	11 (4%)	10 (4%)	21 (4%)
Other	2 (1%)	4 (2%)	6 (1%)
ECOG PS at enrolment			
0	181 (66%)	192 (70%)	373 (68%)
1	85 (31%)	75 (27%)	160 (29%)
2	7 (3%)	8 (3%)	15 (3%)
Bone marrow involvement			
Yes	88 (32%)	91 (33%)	179 (33%)
No	169 (62%)	162 (59%)	331 (60%)
Unknown	3 (1%)	13 (5%)	16 (3%)
Missing	13 (5%)	9 (3%)	22 (4%)
Ann Arbor stage at enrolment			
I/II	52 (19%)	50 (18%)	102 (19%)
III/IV	221 (81%)	225 (82%)	446 (81%)
Met GELF criteria at enrolment	222 (81%)	232 (84%)	454 (83%)
Follicular lymphoma grade			
1/2	203 (74%)	203 (74%)	406 (74%)
3A	67 (25%)	71 (26%)	138 (25%)
B symptoms	63 (23%)	67 (24%)	130 (24%)
FLIPI score at enrolment			
0–1	57 (21%)	57 (21%)	114 (21%)
2	79 (29%)	67 (24%)	146 (27%)
3–5	137 (50%)	150 (55%)	287 (52%)
Missing	0	1 (<1%)	1 (<1%)
Number of previous lines of therapy	1 (1–7)	1 (1–10)	1 (1–10)
1	147 (54%)	153 (56%)	300 (55%)
2	66 (24%)	71 (26%)	137 (25%)
3	39 (14%)	30 (11%)	69 (13%)
≥4	21 (8%)	21 (8%)	42 (8%)

(Table 1 continues on next page)

Outcomes

The primary endpoint was progression-free survival (defined as time from randomisation to first documented disease progression, relapse, or death from any cause), assessed by the local on-site investigator per Lugano 2014 criteria.[25] Key secondary endpoints included investigator-assessed PET-complete response (PET-CR) rate in the FDG-avid population (defined as proportion of patients achieving complete metabolic response at any time after start of treatment per Lugano classification among those with a positive baseline PET scan) assessed by investigator, and overall survival. Additional secondary endpoints were progression-free survival assessed by independent review committee, and overall response rate and duration of response (defined as time from first tumour response [complete or partial response] until time of first documented disease progression, relapse, or death from any cause) assessed by both investigator and independent review committee. Minimal residual disease-negativity rate in peripheral blood and bone marrow (at a threshold of 10^{-5}) was planned for central analysis using an IgH clonality and somatic hyper-mutation next-generation sequencing assay at the end of treatment, as well as quality-of-life and safety assessments as secondary endpoints. Quality of life was evaluated using the 5-level EQ-5D (EQ-5D-5L), the European Organisation for Research and Treatment of Cancer Core Quality of Life Questionnaire-30 items (EORTC QLQ-C30), and Functional Assessment of Cancer Therapy–Lymphoma (FACT-Lym) questionnaires.

The primary efficacy endpoint was further analysed in prespecified subgroups. Exploratory analyses included time to next treatment (defined as time from randomisation to start of next antilymphoma therapy for any reason or death due to any cause), assessed by investigator. A post-hoc exploratory analysis was done to assess progression-free survival in patients who met GELF criteria, assessed by investigator.

Statistical analysis

For the primary analysis, 174 investigator-assessed progression-free survival events in patients with follicular lymphoma were required to detect a hazard ratio (HR) of 0·65 with 80% power, using a two-sided log-rank test at an α-level of 5%. Based on the required number of events, 528 patients with relapsed or refractory follicular lymphoma were to be randomly assigned 1:1 with stratification (n=264 per treatment group), assuming a median progression-free survival of 27·8 months for lenalidomide plus rituximab, 21 months of enrolment, 12 months of follow-up for progression-free survival, and 15% patient dropout.[13] A futility analysis for progression-free survival assessed by the investigator was planned after 20% of required events and a futility boundary of HR ≥1·05, defined by the funder Incyte, was set. At the time of primary analysis, an interim non-binding futility analysis of overall survival was conducted (stop for futility if observed HR >1·24).[26] A final analysis for overall survival is planned at 5 years of follow-up. SAS software (version 9 or later) was used for statistical analyses and to generate all tables and figures.

Efficacy outcomes were assessed in the intention-to-treat population, which comprised all randomised patients, with treatment groups defined according to the assigned treatment at randomisation. Hypothesis testing was formally done on the primary endpoint and the key secondary endpoints and the significance set at 0·05 (two-sided). The primary endpoint analysis served as a gatekeeper and, if the null hypothesis was rejected, the key secondary endpoints were to be tested; hypothesis testing for other endpoints was done for exploratory purposes. Progression-free survival, overall survival, duration of response, and time to next treatment data were analysed using the Kaplan–Meier method and further analysed using a stratified log-rank test. HRs and 95% CIs were estimated using a stratified Cox proportional hazard model. In the progression-free survival analysis, patients with no disease progression, relapse, or death were censored at date of last tumour assessment before data cutoff or new antilymphoma therapy; in the overall survival analysis, patients with no observed death were censored at date last known to be alive. For duration of response, patients with a response but no documented disease progression or death at time of analysis were censored at date of last adequate tumour assessment. For time to next treatment, patients without documented initiation of new antilymphoma therapy or death were censored at date of last contact. PET-CR rate and overall response rate, including 95% CIs, were calculated using the Clopper–Pearson method and were compared between treatment groups using a stratified Cochran–Mantel–Haenszel test, with estimation of odds ratio (OR) with 95% CIs. Best overall response rate was defined as the proportion of patients who achieved a complete response or partial response per the Lugano 2014 criteria any time during the study, before disease progression or next antilymphoma therapy. Partial dates for diagnosis, previous lines of therapy, and adverse events were imputed to derive duration or time to event endpoints; partial death dates were imputed for the overall survival analysis.

For quality-of-life measures, descriptive statistics were tabulated for each of the scores and change from baseline for each score was summarised by visit for each treatment group.

Safety was evaluated in the safety population, which comprised all randomised patients who received at least one dose of study drug according to the actual treatment received, regardless of assigned treatment at the time of randomisation. Minimal residual disease-negativity rate

	Tafasitamab plus lenalidomide plus rituximab (n=273)	Placebo plus lenalidomide plus rituximab (n=275)	Total (N=548)
(Continued from previous page)			
Previous systemic anticancer treatment regimens			
Anthracyclines	161 (59%)	167 (61%)	328 (60%)
Anti-CD20 plus CHOP	151 (55%)	161 (59%)	312 (57%)
Bendamustine alone or in combination	114 (42%)	101 (37%)	215 (39%)
Anti-CD20 monotherapy	54 (20%)	51 (19%)	105 (19%)
Anti-CD20 monotherapy as only previous treatment	26 (10%)	19 (7%)	45 (8%)
Anti-CD20 plus CVP	30 (11%)	43 (16%)	73 (13%)
Chemotherapy plus ASCT	19 (7%)	23 (8%)	42 (8%)
Relapsed or refractory status to last therapy			
Relapsed	148 (54%)	164 (60%)	312 (57%)
Refractory	112 (41%)	97 (35%)	209 (38%)
Undetermined	13 (5%)	14 (5%)	27 (5%)
Refractory to previous anti-CD20	118 (43%)	115 (42%)	233 (43%)
POD24 positive	85 (31%)	88 (32%)	173 (32%)
Time since last antilymphoma therapy			
≤2 years	147 (54%)	157 (57%)	304 (56%)
>2 years	126 (46%)	118 (43%)	244 (45%)
Time since initial follicular lymphoma diagnosis, years	5·2 (0–34)	5·5 (1–33)	5·3 (0–34)
Follicular lymphoma diagnosis confirmed by central pathology	256 (94%)	251 (91%)	507 (93%)

Data are n (%) or median (range). ASCT=autologous stem cell transplantation. CHOP=cyclophosphamide, doxorubicin, vincristine, and prednisone. CVP=cyclophosphamide, vincristine, and prednisone. ECOG PS=Eastern Cooperative Oncology Group performance status. FLIPI=Follicular Lymphoma International Prognostic Index. GELF=Groupe d'Etude des Lymphomes Folliculaires. POD24=progression of disease within 24 months of initial diagnosis.

Table 1: **Baseline patient characteristics (intention-to-treat population)**

is not reported in this manuscript as there were not sufficient samples available at the end of treatment for a meaningful analysis to be done.

Role of the funding source

The funder was responsible for study design and conduct, and data curation and analysis, and participated

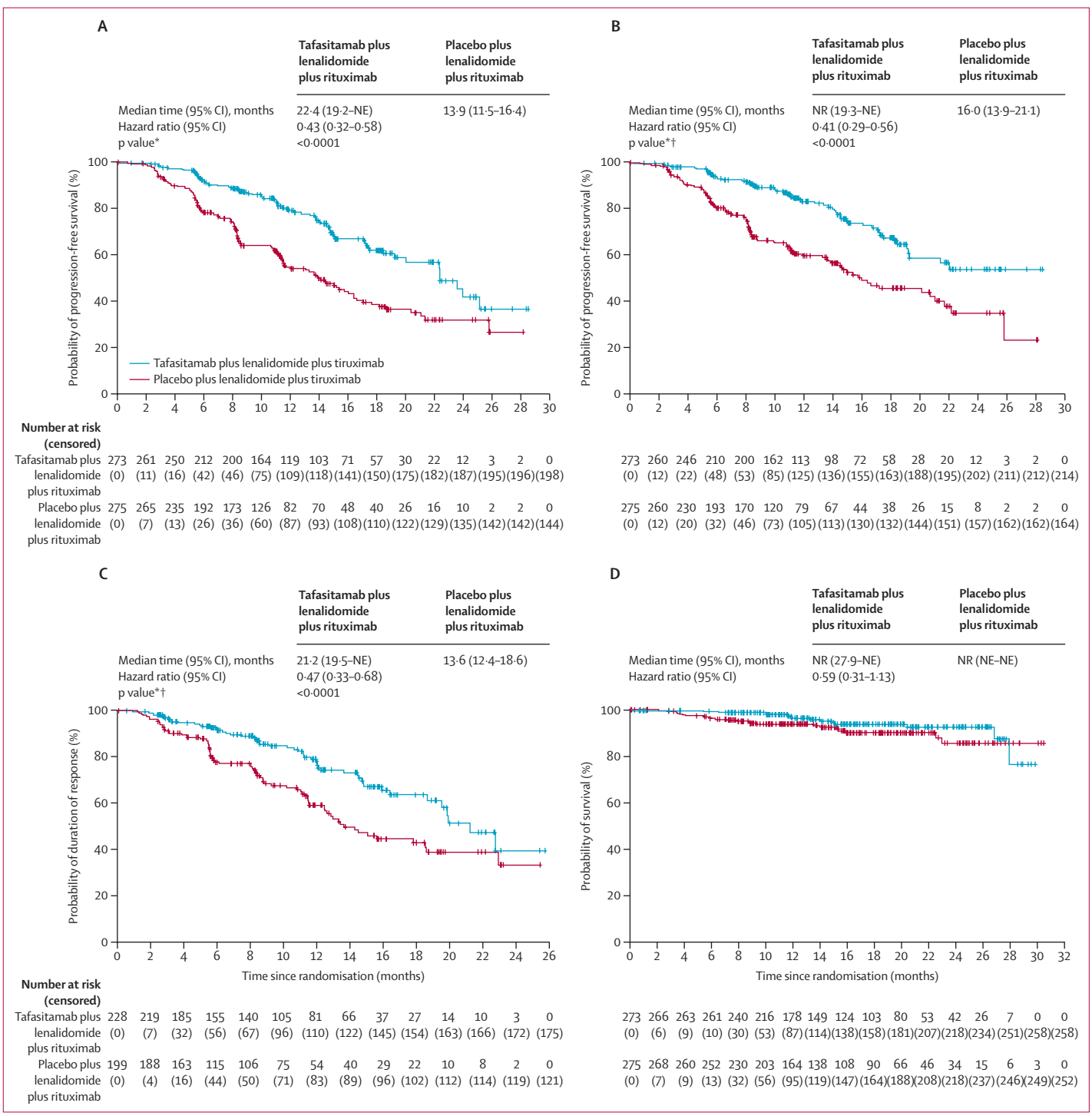

Figure 2: **Progression-free survival (A, B), duration of response (C), and overall survival (D) (intention-to-treat population)**
(A) Kaplan–Meier analysis of progression-free survival as assessed by the investigator. (B) Kaplan–Meier analysis of progression-free survival as assessed by independent review committee, who were masked to treatment assignment. (C) Duration of response in patients who had an overall response (partial or complete response; 228 of 273 patients in the tafasitamab group *vs* 199 of 275 patients in the placebo group). (D) Overall survival. Tick marks indicate censored data. NE=not evaluable. NR=not reached. *Stratified log-rank test with an overall two-sided significance level of 5%. †Nominal p value.

in data interpretation, manuscript preparation, and the decision to submit.

Results

A total of 817 patients were assessed for study eligibility, of whom 654 with relapsed or refractory follicular lymphoma or marginal zone lymphoma were enrolled between April 16, 2021, and Aug 10, 2023. As of the data cutoff date of Feb 23, 2024, 548 patients with follicular lymphoma (intention-to-treat population) had been randomly assigned to receive tafasitamab (n=273) or placebo (n=275); 546 patients were treated (safety population, figure 1). All patients had the indication for treatment established by an investigator, with 222 (81%) of 273 in the tafasitamab group and 232 (84%) of 275 in the placebo group having met at least one GELF criterion. Baseline demographics and disease characteristics were similar between treatment groups (table 1), with a median age of 64 years (range 31–88). Of 548 patients assigned to treatment, 433 (79%) had an intermediate-risk or high-risk Follicular Lymphoma International Prognostic Index score, median number of lines of previous therapy was 1 (range 1–10), 248 (45%) had received at least two previous lines of therapy, 173 (32%) had POD24, 209 (38%) were refractory to their last previous therapy, and 233 (43%) were refractory to a previous anti-CD20 monoclonal antibody. Concordance between local and central pathology for diagnosis of

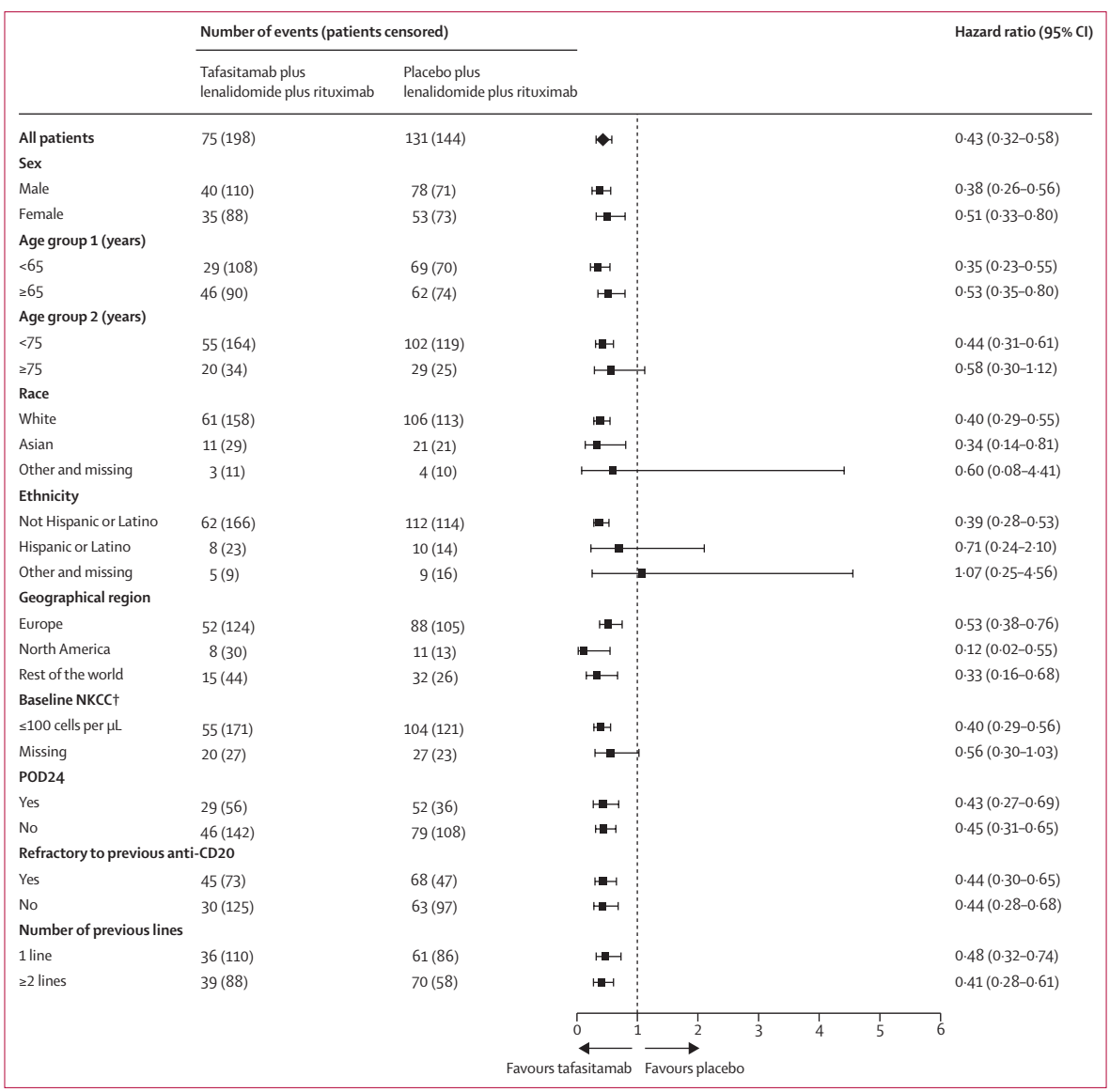

Figure 3: **Forest plot of hazard ratios for progression-free survival (intention-to-treat population)***
NKCC=natural killer cell count. POD24=progression of disease within 24 months of initial diagnosis. *Subgroup analyses are based on stratification factor.
†No patient had baseline NKCC >100 cells per µL.

	Tafasitamab plus lenalidomide plus rituximab	Placebo plus lenalidomide plus rituximab
Overall response rate (intention-to-treat population)		
Patients	273	275
Best response*		
Complete response	124 (45%)	94 (34%)
Partial response	104 (38%)	105 (38%)
Stable disease	28 (10%)	46 (17%)
Progressive disease	7 (3%)	20 (7%)
Not evaluable	2 (1%)	0
Not done	8 (3%)	10 (4%)
Complete response (radiographic)†	141 (52%)	110 (40%)
Overall response rate‡, % (95% CI)§	84% (78·6–87·7)	72% (66·7–77·6)
Odds ratio¶||| (95% CI)§	2·0 (1·30–3·02)	..
Nominal p value||	0·0014	..
PET-CR (FDG-avid population)		
Patients with FDG-avid positive PET scan at baseline	251	254
Patients with post-baseline PET assessments**	201/251 (80%)	205/254 (81%)
Best metabolic response based on PET		
Complete metabolic response (PET-CR)	124 (49%)	101 (40%)
Partial metabolic response	37 (15%)	39 (15%)
No metabolic response/stable disease	19 (8%)	12 (5%)
Progressive metabolic disease	19 (8%)	51 (20%)
PET after confirmed progressive disease or new anti-lymphoma treatment initiation	2 (1%)	2 (1%)
Not done	50 (20%)	49 (19%)
PET-CR††, % (95% CI)§	49% (43·1–55·8)	40% (33·7–46·1)
Odds ratio¶|| (95% CI)§	1·5 (1·04–2·13)	..
p value||	0·029	..

Data are n (%) or n/N (%) unless otherwise specified. CR=complete response. FDG=fluorodeoxyglucose. *Per Lugano 2014 criteria. †Complete response (radiographic) included patients with positive baseline bone marrow morphological involvement without confirmed morphological complete response (bone marrow not done or not evaluable) after radiographic complete response. ‡Overall response rate was defined as the proportion of patients who achieved a complete or partial response as determined per Lugano classification at any time during the study but before the first progressive disease and before or at the start of a new antilymphoma treatment. §Calculated using Clopper–Pearson method. ¶The strata information was based on the data obtained from the interactive response technology used for randomisation. ||By stratified Cochran–Mantel–Haenszel. **Percentages are calculated based on patients with a positive PET scan at baseline, defined as having a Deauville score of 4 or 5 at baseline. ††The PET-CR rate is defined as the proportion of patients who achieved complete metabolic response at any time after start of treatment as per Lugano criteria in patients with FDG-avid disease at baseline by PET-CR results only. Patients with no post-baseline assessment by PET, or who did not achieve a complete metabolic response, were not considered PET-CR responders.

Table 2: **Best overall response (intention-to-treat population) and PET-CR (FDG-avid population) as assessed by the investigator**

follicular lymphoma was 93% (507 of 548) overall: 94% (256 of 273) in patients receiving tafasitamab and 91% (251 of 275) in patients receiving placebo. Details regarding previous therapies received by patients can be found in table 1. At data cutoff, 51 (19%) of 273 patients receiving tafasitamab and 42 (15%) of 275 receiving placebo were still on treatment; 222 (81%) patients receiving tafasitamab and 231 (84%) receiving placebo had discontinued treatment (figure 1).

After a median follow-up of 14·1 months (14·3 months [95% CI 11·8–15·0] in the tafasitamab treatment group; 14·1 months [11·5–15·0] in the placebo treatment group), the primary endpoint of investigator-assessed progression-free survival was found to be significantly superior with tafasitamab plus lenalidomide and rituximab (HR 0·43 [95% CI 0·32–0·58], p<0·0001; figure 2A), resulting in a significant reduction in the risk of progression, relapse, or death. Median progression-free survival was 22·4 months (95% CI 19·2 to not evaluable [NE]) in the tafasitamab treatment group versus 13·9 months (11·5–16·4) in the placebo treatment group. The primary endpoint was supported when assessed by independent review committee masked to study treatment, with progression-free survival shown to be significantly longer with tafasitamab versus placebo (median not reached [95% CI 19·3–NE] *vs* 16·0 months [13·9–21·1]; HR 0·41 [95% CI 0·29–0·56], p<0·0001; figure 2B). A consistent benefit in progression-free survival with tafasitamab was observed versus placebo for all prespecified subgroups, including patients who were refractory to an anti-CD20 monoclonal antibody, those who had received multiple previous lines of therapy, and regardless of geographical region (figure 3 and appendix p 10). Tafasitamab was associated with a longer median progression-free survival versus placebo regardless of POD24 status (19·2 months [95% CI 13·8–NE] *vs* 11·3 months [8·3–13·6] for POD24-positive patients and 23·6 months [22·3–NE] *vs* 16·0 months [13·3–21·4] for POD24-negative patients; appendix p 10). In a post-hoc exploratory analysis in the subgroup of patients who met GELF criteria, median progression-free survival was longer with tafasitamab versus placebo (22·4 months [95% CI 17·5–NE] *vs* 13·6 months [11·5–16·4]; HR 0·46 [95% CI 0·34–0·63]), consistent with results for the intention-to-treat population.

Among the patients who had FDG-avid disease at baseline, complete metabolic response was achieved by 124 of 251 patients in the tafasitamab group and 101 of 254 patients in the placebo group, resulting in a significantly higher PET-CR rate of 49% versus 40% (OR 1·5 [95% CI 1·04–2·13], p=0·029; table 2); 37 (15%) patients in the tafasitamab group and 39 (15%) patients in the placebo group achieved partial metabolic response, whereas 19 (8%) patients in the tafasitamab group and 51 (20%) patients in the placebo group had progressive metabolic disease.

Best overall response rate was assessed in the intention-to-treat population and was 84% (95% CI 78·6–87·7) with tafasitamab and 72% (66·7–77·6) with placebo (OR 2·0 [95% CI 1·30–3·02], p=0·0014; table 2). Median duration of response was 21·2 months (95% CI 19·5–NE) with tafasitamab and 13·6 months (12·4–18·6) with placebo (HR 0·47 [95% CI 0·33–0·68], p<0·0001; figure 2C). Time to next treatment was longer with tafasitamab versus placebo (median not reached [NE–NE] with tafasitamab *vs* 28·8 [20·7–NE] months with placebo; HR 0·45 [95% CI 0·31–0·64], p<0·0001), with 226 (83%) of 273 patients receiving tafasitamab versus 186 (68%) of 275 receiving placebo alive and not having started a new antilymphoma treatment at data cutoff (appendix p 12).

	Tafasitamab plus lenalidomide plus rituximab (n=274*)		Placebo plus lenalidomide plus rituximab (n=272†)		Total (N=546)	
	Grade 1–2	Grade 3–4	Grade 1–2	Grade 3–4	Grade 1–2	Grade 3–4
Any adverse event	268 (98%)	195 (71%)	263 (97%)	189 (69%)	531 (97%)	384 (70%)
Diarrhoea	103 (38%)	2 (1%)	77 (28%)	5 (2%)	180 (33%)	7 (1%)
Constipation	79 (29%)	2 (1%)	67 (25%)	0	146 (27%)	2 (<1%)
Neutropenia	78 (29%)	109 (40%)	59 (22%)	102 (38%)	137 (25%)	211 (39%)
COVID-19	76 (28%)	16 (6%)	60 (22%)	6 (2%)	136 (25%)	22 (4%)
Rash	60 (22%)	3 (1%)	58 (21%)	1 (<1%)	118 (22%)	4 (1%)
Fatigue	57 (21%)	5 (2%)	43 (16%)	0	100 (18%)	5 (1%)
Cough	52 (19%)	0	47 (17%)	0	99 (18%)	0
Muscle spasms	49 (18%)	0	49 (18%)	0	98 (18%)	0
Pyrexia	50 (18%)	4 (1%)	40 (15%)	6 (2%)	90 (16%)	10 (2%)
Nausea	48 (18%)	1 (<1%)	37 (14%)	1 (<1%)	85 (16%)	2 (<1%)
Infusion-related reaction	42 (15%)	2 (1%)	40 (15%)	1 (<1%)	82 (15%)	3 (1%)
Pruritus	44 (16%)	1 (<1%)	28 (10%)	0	72 (13%)	1 (<1%)
Thrombocytopenia	32 (12%)	17 (6%)	36 (13%)	20 (7%)	68 (12%)	37 (7%)
Anaemia	31 (11%)	12 (4%)	32 (12%)	16 (6%)	63 (12%)	28 (5%)
Asthenia	36 (13%)	3 (1%)	27 (10%)	2 (1%)	63 (12%)	5 (1%)
Oedema peripheral	20 (7%)	0	35 (13%)	3 (1%)	55 (10%)	3 (1%)
Upper respiratory tract infection	25 (9%)	1 (<1%)	29 (11%)	0	54 (10%)	1 (<1%)
Hypokalaemia	22 (8%)	4 (1%)	30 (11%)	3 (1%)	52 (10%)	7 (1%)
Decreased appetite	27 (10%)	0	23 (8%)	2 (1%)	50 (9%)	2 (<1%)
Back pain	31 (11%)	0	15 (6%)	0	46 (8%)	0
Abdominal pain	20 (7%)	0	25 (9%)	5 (2%)	45 (8%)	5 (1%)
Headache	27 (10%)	1 (<1%)	18 (7%)	0	45 (8%)	1 (<1%)
Respiratory tract infection	18 (7%)	2 (1%)	24 (9%)	1 (<1%)	42 (8%)	3 (1%)
Dizziness	21 (8%)	0	19 (7%)	0	40 (7%)	0
Arthralgia	19 (7%)	0	21 (8%)	0	40 (7%)	0
Dyspnoea	18 (7%)	1 (<1%)	21 (8%)	5 (2%)	39 (7%)	6 (1%)
Insomnia	21 (8%)	0	17 (6%)	0	38 (7%)	0
Nasopharyngitis	17 (6%)	0	20 (7%)	0	37 (7%)	0
Increased alanine aminotransferase	18 (7%)	2 (1%)	18 (7%)	2 (1%)	36 (7%)	4 (1%)
Oropharyngeal pain	24 (9%)	1 (<1%)	10 (4%)	0	34 (6%)	1 (<1%)
Urinary tract infection	18 (7%)	2 (1%)	15 (6%)	3 (1%)	33 (6%)	5 (1%)
Rash maculopapular	18 (7%)	3 (1%)	14 (5%)	1 (<1%)	32 (6%)	4 (1%)
Vomiting	19 (7%)	0	12 (4%)	2 (1%)	31 (6%)	2 (<1%)
Pneumonia	17 (6%)	23 (8%)	13 (5%)	14 (5%)	30 (5%)	37 (7%)
Decreased neutrophils	17 (6%)	16 (6%)	12 (4%)	18 (7%)	29 (5%)	34 (6%)
Pain in extremity	21 (8%)	0	7 (3%)	0	28 (5%)	0
Myalgia	15 (6%)	0	13 (5%)	1 (<1%)	28 (5%)	1 (<1%)
Decreased weight	13 (5%)	0	14 (5%)	0	27 (5%)	0
Chills	13 (5%)	0	13 (5%)	0	26 (5%)	0
Dyspepsia	14 (5%)	0	12 (4%)	0	26 (5%)	0
Increased aspartate aminotransferase	13 (5%)	1 (<1%)	13 (5%)	0	26 (5%)	1 (<1%)
Decreased white blood cell count	13 (5%)	5 (2%)	12 (4%)	3 (1%)	25 (5%)	8 (1%)
Rhinorrhoea	14 (5%)	0	10 (4%)	0	24 (4%)	0
Influenza	15 (6%)	0	8 (3%)	1 (<1%)	23 (4%)	1 (<1%)
Hypomagnesaemia	13 (5%)	1 (<1%)	8 (3%)	0	21 (4%)	1 (<1%)
Paraesthesia	13 (5%)	0	9 (3%)	0	22 (4%)	0
Neuropathy peripheral	13 (5%)	0	6 (2%)	0	19 (3%)	0
Hyperglycaemia	7 (3%)	4 (1%)	7 (3%)	1 (<1%)	14 (3%)	5 (1%)

(Table 3 continues on next page)

	Tafasitamab plus lenalidomide plus rituximab (n=274*)		Placebo plus lenalidomide plus rituximab (n=272†)		Total (N=546)	
	Grade 1–2	Grade 3–4	Grade 1–2	Grade 3–4	Grade 1–2	Grade 3–4
(Continued from previous page)						
Hypertension	6 (2%)	1 (<1%)	7 (3%)	4 (1%)	13 (2%)	5 (1%)
Pulmonary embolism	2 (1%)	4 (1%)	1 (<1%)	3 (1%)	3 (1%)	7 (1%)
COVID-19 pneumonia	3 (1%)	13 (5%)	0	3 (1%)	3 (1%)	16 (3%)
Acute kidney injury	1 (<1%)	8 (3%)	1 (<1%)	6 (2%)	2 (<1%)	14 (3%)
Febrile neutropenia	0	12 (4%)	1 (<1%)	6 (2%)	1 (<1%)	18 (3%)
Syncope	0	4 (1%)	0	2 (1%)	0	6 (1%)

Data are n (%); patients were counted once under each MedDRA preferred term, for grade 1–2 and for grade 3–4 adverse events. Cutoffs were based on clinical relevance to assess the safety profile of the treatments. MedDRA=Medical Dictionary for Regulatory Activities. *One patient randomly assigned to the placebo plus lenalidomide and rituximab group is included in the tafasitamab plus lenalidomide and rituximab safety population because the patient erroneously received tafasitamab. †Three patients randomly assigned to the placebo plus lenalidomide and rituximab group are not included in the safety population because they erroneously received tafasitamab (n=1) or did not receive any study treatment due to confirmation of rituximab hypersensitivity (n=1), and withdrawal by patient (n=1).

Table 3: **Most frequent grade 1–2 (≥5% in any group) and grade 3–4 (≥2% in any group) adverse events in the safety population**

	Tafasitamab plus lenalidomide plus rituximab (n=274*)	Placebo plus lenalidomide plus rituximab (n=272†)	Total (N=546)
Any serious adverse event	99 (36%)	86 (32%)	185 (34%)
Pneumonia	21 (8%)	13 (5%)	34 (6%)
COVID-19	19 (7%)	7 (3%)	26 (5%)
COVID-19 pneumonia	14 (5%)	5 (2%)	19 (3%)
Acute kidney injury	8 (3%)	4 (1%)	12 (2%)
Febrile neutropenia	7 (3%)	6 (2%)	13 (2%)
Pyrexia	4 (1%)	7 (3%)	11 (2%)
Sepsis	3 (1%)	3 (1%)	6 (1%)
Neutropenia	2 (1%)	3 (1%)	5 (1%)
Abdominal pain	0	5 (2%)	5 (1%)
Diarrhoea	0	3 (1%)	3 (<1%)

Data are n (%); patients were counted once under each MedDRA preferred term. Cutoff is based on clinical relevance to assess the safety profile of the treatments. MedDRA=Medical Dictionary for Regulatory Activities. *One patient randomly assigned to the placebo plus lenalidomide and rituximab group is included in the tafasitamab plus lenalidomide and rituximab safety population because the patient erroneously received tafasitamab. †Three patients randomly assigned to the placebo plus lenalidomide and rituximab group are not included in the safety population because they erroneously received tafasitamab (n=1) or did not receive any study treatment due to confirmation of rituximab hypersensitivity (n=1), and withdrawal by patient (n=1).

Table 4: **Most frequent serious adverse events (≥1% in any group) in the safety population**

For the overall survival futility analysis, with median follow-up of 15·3 months (15·8 months [95% CI 14·6–16·9] for the tafasitamab treatment group; 14·6 months [13·0–15·9] for the placebo treatment group) and 38 deaths (15 in the tafasitamab treatment group, 23 in the placebo treatment group), the preliminary data are immature with regard to the difference between tafasitamab versus placebo (HR 0·59 [95% CI 0·31–1·13], figure 2D). A full analysis of the overall survival is planned at 5 years of follow-up.

Quality of life, as measured by the EQ-5D-5L scale, the EORTC QLQ-C30 scale, and the five FACT-Lym subscales and three composite scores was similar in the two treatment groups (data not shown).

The median number of treatment cycles received by patients was similar between groups (tafasitamab 12 cycles [range 1–12]; placebo 11 cycles [range 1–12]), and exposure to lenalidomide and rituximab was also similar in the tafasitamab and placebo groups (appendix p 6). Overall, within the safety population, 99% (272 of 274) of patients receiving tafasitamab and 99% (270 of 272) receiving placebo had an adverse event (appendix p 6). Most common any-grade adverse events occurring in at least 20% of patients in either treatment group were neutropenia (133 [49%] in the tafasitamab group *vs* 123 [45%] in the placebo group), diarrhoea (103 [38%] *vs* 77 [28%]), COVID-19 (86 [31%] *vs* 64 [24%]), constipation (80 [29%] *vs* 67 [25%]), rash (60 [22%] *vs* 58 [21%]), and fatigue (58 [21%] *vs* 43 [16%]; appendix p 8). Table 3 summarises grade 1–2 and grade 3–4 adverse events reported during the study. There were no notable differences in mean IgG, IgA, or IgM concentrations at baseline and no meaningful changes were observed during the study among patients receiving tafasitamab or placebo (appendix p 13). Concomitant immunoglobulins were administered to 37 (14%) of 274 patients receiving tafasitamab and 23 (8%) of 272 receiving placebo based on standard protocols at the investigator's site.

Incidence of grade 3 or 4 adverse events and serious adverse events was similar with tafasitamab and placebo (tables 3 and 4). Most common grade 3 or 4 adverse events occurring in patients receiving tafasitamab or placebo included neutropenia, pneumonia, thrombocytopenia, decreased neutrophils, COVID-19, COVID-19 pneumonia, and anaemia (table 3). The most frequent

serious adverse events occurring in patients receiving tafasitamab or placebo included pneumonia, COVID-19, and COVID-19 pneumonia (table 4).

Dose interruptions and treatment discontinuations due to adverse events were similar between tafasitamab and placebo (appendix p 6); 203 (74%) of 274 patients receiving tafasitamab and 190 (70%) of 272 receiving placebo had dose delay or interruption due to adverse events, and 30 (11%) patients receiving tafasitamab and 18 (7%) receiving placebo discontinued treatment due to adverse events.

A total of 15 (6%) of 274 patients receiving tafasitamab and 23 (9%) of 272 receiving placebo died during the study. Five (2%) patients receiving tafasitamab and 17 (6%) receiving placebo died due to disease progression. Six (2%) patients receiving tafasitamab and six (2%) receiving placebo died due to fatal adverse events, most commonly due to infection (appendix p 9). No patient receiving tafasitamab had a fatal adverse event related to treatment; two (1%) patients receiving placebo had fatal adverse events related to treatment.

Discussion

In the phase 3 inMIND study, the addition of tafasitamab to lenalidomide plus rituximab was associated with longer progression-free survival (median 22·4 months *vs* 13·9 months) than lenalidomide plus rituximab in patients with relapsed or refractory follicular lymphoma, representing a significant reduction in risk of progression, relapse, or death. Addition of tafasitamab also resulted in a higher overall response rate, including complete response rate, and longer duration of response. Patients treated with tafasitamab also had a longer time to next antilymphoma treatment than the placebo group. Overall survival analysis is immature and differences observed between tafasitamab and placebo will be fully evaluated at the time of formal testing planned after 5 years of follow-up.

Addition of tafasitamab to lenalidomide and rituximab was generally well tolerated. Similar rates of overall adverse events were reported in patients treated with tafasitamab or placebo, including similar rates of grade 3 or 4 adverse events and serious adverse events. Types of observed adverse events were consistent with known safety profiles of the trial treatments, although rates of diarrhoea, COVID-19, and pneumonia were slightly numerically higher in the tafasitamab group compared with the placebo group. In general, addition of tafasitamab did not compromise delivery of lenalidomide or rituximab. The incidence of COVID-19 infections was 31% with tafasitamab compared with 24% with placebo, which might reflect additional immunosuppression introduced by anti-CD19 directed therapy. However, it should be noted that the study was done during the peak of the COVID-19 pandemic, and very few patients had received COVID-19 vaccination before screening (only 5% [14 of 273] of patients in the tafasitamab group and 4% [12 of 275] of patients in the placebo group).

First-line chemo-immunotherapy for follicular lymphoma is associated with prolonged progression-free survival, although approximately 20% of patients have POD24 and have poorer outcomes than those without POD24.[3–6,27,28] There is no standard of care for patients with relapsed or refractory follicular lymphoma, for whom prognosis generally worsens with each subsequent therapy.[10] The goal of therapy has been to prolong remission, while maintaining quality of life, making immunotherapy an attractive option. Since 2021, chimeric antigen receptor therapy and bispecific antibodies have been approved for follicular lymphoma.[29–38] Given the paucity of alternatives in the current therapeutic landscape, tafasitamab with lenalidomide plus rituximab offers another second-line treatment option.

However, there is a clear need to further expand therapeutic options for patients with relapsed or refractory follicular lymphoma and numerous novel combinations are under evaluation. In 2025, a single-arm, phase 2 trial of the anti-CD19 antibody–drug conjugate loncastuximab tesirine in combination with rituximab in patients with relapsed or refractory follicular lymphoma showed a promising complete response rate of 67% at week 12 of treatment.[39] A phase 2, single-arm trial of the CD3/CD20 bispecific antibody mosunetuzumab as monotherapy reported an encouraging complete response rate of 60%.[29] Interim analysis of the phase 3 EPCORE FL-1 trial evaluating the CD3/CD20 bispecific antibody epcoritamab plus lenalidomide and rituximab versus lenalidomide and rituximab alone in adult patients with relapsed or refractory follicular lymphoma indicated that the study met one of its dual primary endpoints of improved objective response rate.[40] Comparison of these published data with the data presented in our study is difficult, due to likely differences in patient populations. Treatment options for this patient population are associated with varying degrees of toxicity and availability; it is likely that the choice among treatment options will require an individualised approach, balancing risk and benefit, accessibility, and patient preferences.

Lenalidomide and rituximab has proven efficacy in both first-line and relapsed or refractory follicular lymphoma,[13,41,42] and is currently approved following at least one previous line of therapy based on data from AUGMENT, which compared lenalidomide and rituximab versus rituximab alone in patients not refractory to rituximab. Efficacy of lenalidomide and rituximab, as measured by median progression-free survival by investigator in our study (13·9 months), was lower than previously reported in AUGMENT (27·8 months).[13] However, it is important to note that inMIND included a patient cohort at higher risk than AUGMENT, more typical of patients treated in clinical practice; 83% of the inMIND population had high tumour burden based on

GELF criteria versus 51% in AUGMENT, 52% of the inMIND population had a high (≥3) Follicular Lymphoma International Prognostic Index score at study entry versus 34% in AUGMENT, 39% of patients in inMIND had previous exposure to bendamustine alone or in combination versus 9% of patients who had previously received rituximab plus bendamustine in AUGMENT, and 43% of the inMIND population was refractory to anti-CD20 monoclonal antibody therapy, whereas the AUGMENT trial excluded patients with rituximab-refractory disease.[13] Interim results from the induction phase of the MAGNIFY study, designed to examine the optimal duration of lenalidomide therapy in patients with relapsed or refractory follicular or marginal zone lymphoma, showed a median progression-free survival by investigator of 50·5 months and an overall response rate of 74%.[43] However, differences in study design and patient characteristics of MAGNIFY and inMIND might make future comparisons difficult once final trial results are available. Currently, there is a lack of large, representative, real-world studies evaluating the efficacy of lenalidomide and rituximab in patients with relapsed or refractory follicular lymphoma. A single-centre, real-world retrospective study of 69 patients, published in 2025, showed a median progression-free survival of 25 months following lenalidomide and rituximab.[44] However, the small sample size and potential patient selection biases limit interpretation.

The findings of our study provide support for the efficacy of the addition of tafasitamab to lenalidomide and rituximab, with benefit seen in all prespecified subgroups analysed. In particular, benefit was seen regardless of POD24 status, anti-CD20 refractoriness, or number of previous lines of therapy. Overall survival data are immature and will be fully evaluated at the planned 5-year analysis to provide more robust data on this outcome. Future work will also examine CD19 expression in patients following treatment with tafasitamab, providing insight on optimal subsequent lines of therapy. One limitation of the study is that minimal residual disease-negativity rate at the end of treatment could not be fully evaluated due to the small number of samples available for analysis. A second potential limitation is that assessment of morphological complete response in several patients with positive baseline bone marrow involvement could not be evaluated or was not performed by data cutoff. Although bone marrow biopsies have typically been recommended in clinical trials, particularly for patients without bone marrow uptake on FDG PET–CT at baseline, evidence suggests bone marrow biopsies do not affect response assessment in follicular lymphoma clinical trials.[45–48]

In conclusion, the phase 3 inMIND study evaluating tafasitamab plus lenalidomide and rituximab in patients with relapsed or refractory follicular lymphoma showed statistically significant and clinically meaningful improvement in progression-free survival, overall and complete response rate, and duration of response versus patients receiving placebo plus lenalidomide and rituximab. The safety profile was manageable and consistent with expectations for the patient population. This study is the first to validate the approach of combining two unconjugated monoclonal antibodies (anti-CD19 with anti-CD20) in treatment of lymphoma in a large, phase 3 randomised trial. Tafasitamab, lenalidomide, and rituximab can be administered in community as well as academic settings, and represents a potential new standard-of-care treatment for relapsed or refractory follicular lymphoma.

Contributors

LHS, KH, SL, CWS, AS, SP, BEW, PP, H-PL, AJ-U, J-MS, TMK, EDD, TK, CP, CT, DD, and MT were responsible for recruitment of patients, trial conduct, and data collection. LHS, EdW, MA, and MT contributed to the development and drafting of the manuscript. All authors contributed to the interpretation of the data, critical review of the manuscript, and provided approval of the final version to be published. All authors had full access to all the data in the study and final responsibility for the decision to submit for publication. LHS, EdW, MA, and IV had direct access and verified the underlying data reported in the manuscript.

Declaration of interests

LHS participated in advisory boards for AbbVie, Acerta Pharma, Apobiologix, AstraZeneca, Celgene, Debiopharm, Genentech, Genmab, Gilead Sciences, Incyte, Janssen, Karyopharm Therapeutics, Kite Pharma, Lundbeck, Merck, MorphoSys, Novartis, Sandoz, Takeda, TG Therapeutics, and Verastem Oncology; participated in an advisory board for and received research grants from Teva; and participated in advisory board and speakers bureaus for Roche and Seattle Genetics. KH participated in advisory boards and speakers bureaus for AbbVie, BeiGene, Bristol-Myers Squibb, EUSA Pharma/Recordati, Gilead Sciences, Incyte, Novartis, Roche, and Servier. SL participated in advisory boards for AbbVie, Bristol-Myers Squibb, Genmab, Incyte, Janssen, Kite Pharma, Novartis, Regeneron, and Roche. CWS received honoraria from AstraZeneca, Gilead Sciences, Janssen, Pfizer, and Roche; and participated in advisory boards for Bristol-Myers Squibb, Celgene, Daiichi Sankyo, Gilead Sciences, Hexal, Incyte, Janssen, Merck Serono, Novartis, Roche, and Takeda. AS reports a consultant role and participation in speakers bureaus for AbbVie, AstraZeneca, BeiGene, Incyte, Ipsen, Roche, and Sandoz; and received a research grant from Gilead Sciences. SP received honoraria from AbbVie, BeiGene, Celgene, Gilead Sciences, Janssen, and Roche. BEW reports a consultant role for Roche; received research grants from Gilead Sciences; and received honoraria from Incyte and MorphoSys. PP participated in advisory boards for AbbVie, AstraZeneca, Bristol-Myers Squibb, Janssen, Novartis, and Roche; and received research grants from AbbVie and AstraZeneca. H-PL received honoraria from Roche; and participated in advisory boards for AstraZeneca, BeiGene, and Gilead Sciences. AJ-U participated in advisory boards for AbbVie, AstraZeneca, Genmab, Kite Pharma, and Lilly; and was a speaker for AbbVie, Incyte, Janssen, Kite Pharma, Novartis, and Roche. J-MS received honoraria from AbbVie, BeiGene, Bristol-Myers Squibb/Celgene, Gilead Sciences/Kite Pharma, Incyte, Janssen, Lilly, and Roche; and participated in advisory boards for AbbVie, AstraZeneca, BeiGene, Bristol-Myers Squibb/Celgene, Gilead Sciences/Kite Pharma, Incyte Corporation, Janssen, Lilly, Miltenyi Biomedicine, Novartis, Roche, and Sobi. TMK reports consulting or advisory roles for Amgen, AstraZeneca/MedImmune, Boryung, Daiichi Sankyo, HK inno.N, IMBdx, Janssen, Novartis, Regeneron, Roche/Genentech, Samsung Bioepis, Takeda, and Yuhan. EDD participated in advisory boards and speakers bureaus for BeiGene, Bristol-Myers Squibb/Celgene, Ideogen, and Takeda. TK received honoraria from AbbVie, AstraZeneca, BeiGene, Genmab, Janssen, and Ono Pharmaceutical. CP participated in advisory boards for Acrotech Biopharma, AstraZeneca, Ipsen, and Seagen; and received research funding from Astex, Dren Bio, Incyte, and Seagen. CT had a speaker role for AbbVie, Amgen, Bayer, Bristol-Myers Squibb/Celgene, Gilead Sciences, Incyte, Janssen, Kite Pharma, Novartis, Roche, and Takeda;

received support for travel and accommodation from AbbVie, Amgen, Bayer, Bristol-Myers Squibb/Celgene, Gilead Sciences, Janssen, Kite Pharma, Novartis, Roche, and Takeda; received research grants from Janssen and Roche; was a consultant for and received honoraria from AbbVie, Amgen, Bayer, Bristol-Myers Squibb/Celgene, Gilead Sciences, Incyte, Janssen, Kite Pharma, Novartis, Roche, and Takeda. DD was a consultant for Alexion, Bristol-Myers Squibb, Incyte, Novartis, Sanofi, Sobi, and Takeda; and received research funding from Alexion, Amgen, Novartis, Roche, and Sobi. EdW, MA, and IV report employment and stock ownership from Incyte. MT received honoraria from AbbVie, Amgen, Bristol-Myers Squibb, Gilead Sciences, Incyte, Janssen, MorphoSys, Roche, and Takeda; was a consultant and participated in advisory boards for AbbVie, Amgen, Bristol-Myers Squibb, Celgene, Gilead Sciences, Incyte, Janssen, MorphoSys, Roche, and Takeda; and received support for travel and accommodation, and expenses from AbbVie, Bristol-Myers Squibb, Gilead Sciences, Janssen, Roche, and Takeda.

Data sharing

Incyte is committed to data sharing that advances science and medicine while protecting patient privacy. Qualified external scientific researchers may request anonymised datasets owned by Incyte for the purpose of conducting legitimate scientific research. Researchers may request anonymised datasets from any interventional study (except phase 1 studies) for which the product and indication have been approved on or after Jan 1, 2020, in at least one major market (eg, USA, Europe, Japan). Data will be available for request after the primary publication or 2 years after the study has ended. Information on Incyte's clinical trial data sharing policy and instructions for submitting clinical trial data requests are available at: https://www.incyte.com/Portals/0/Assets/Compliance%20and%20Transparency/clinical-trial-data-sharing.pdf?ver=2020-05-21-132838-960.

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